Mussolini and ChatGPT. Examining the Risks of AI writing Historical Narratives on Fascism

Lacriola, Michele
michele.lacriola@unisi.it
Università di Siena, Italy

De Ninno, Fabio
fabio.deninno@unisi.it
Università di Siena, Italy

This article is one of the first attempts to analyse the risks of historical content generated using artificial intelligence with Mussolini's Fascist regime as its subject. We chose to test the most prominent AI of our time, Chat GPT, which today stands out for its high performance and widespread use, with OpenAI's website seeing 1.6 billion visits and 180 million users monthly (Crawford, 2021). The authors tested GPT on Italian Fascism, where their academic background lies, finding possible errors or troublesome interpretations when GPT is used as a search engine for historical information.

The importance of this issue cannot be understated. Artificial intelligence is integral to the world's socioeconomic structure, even before generative tools like Chat GPT were available to the public. Since 2015, there have been 2.4 times as many academic publications on AI, with 81.07per cent coming from the education and research sectors (Human-Centered Artificial Intelligence, 2024). AI is also entering students' national curricula, most notably in China, raising questions about its role in STEM (Shen, Yang, et al., 2023).

Nevertheless, the spread of generative intelligence has also affected the humanities. Using such tools for study and academic preparation is now common among students (Črček, Patekar, 2023; AA.VV. 2023; Chan, Hu, 2023; Firat, 2023; Hollfloresand, Ciachir, 2024). AI tools support humanities scholars in managing information, formulating research questions, and analysing data. For historians, AI can transcribe documents, reveal connections between events, track linguistic trends, and reconstruct past contexts using incomplete data (Meadows, Sternfeld, 2023).

Until the rise of Large Language Models (LLM), like ChatGPT, Computers assisted in preserving and organising digital sources, but historians remained reliant on established methodologies. However, LLM are introducing the possibility of computational processes that could generate autonomous reconstructions of past events, prompting the question: 'can a machine document the past?' (Spina, 2024). Sternfeld suggests that AI could be seen as a historian, as its learning process involves collecting, classifying, analysing, interpreting, and preserving historical data. He argues that these tasks require historical awareness, including memory, critical analysis, and causality (Sternfeld, 2023), but this view risks oversimplifying historical research. Historians interpret events within complex cultural, social, and political contexts that AI lacks. In contrast, Leme Lopes critiques Chat GPT's approach to history, noting its answers are often generic, outdated, and lack deep critical analysis. Additionally, bibliographic errors and difficulties distinguishing reliable sources limit its role to a supportive tool rather than a primary resource for historical research (Leme Lopes, 2023).

However, the use of AI in interpreting data analysis results is controversial, as it assumes some metacognitive capability (Chalmers, 2023). Nevertheless, many argue that AI models, described by Emily Benderas as 'stochastic parrots' (Bender, Gebru, McMillan-Major, Shmitchell, 2021), merely replicate statistical patterns without proper understanding, as demonstrated by Marcus and Davis (Marcus, Davis, 2020). This limits the quality and originality of AI-generated text and its ability to handle unfamiliar contexts. While AI cannot match a historian's interpretive skills, it can still help establish shared conceptual and methodological ground among researchers, potentially boosting scientific productivity (Noy, Zhang, 2023).

While Chat GPT offers vast possibilities, it also presents challenges that need to be addressed. The accuracy of its results depends heavily on the quality, diversity, and complexity of its training data and the input provided by users. Natural Language Processing systems may struggle with the nuances of human language, leading to misunderstandings. Additionally, biased or incomplete training data can introduce inaccuracies (Schramowski, Turan, Andersen et al., 2022). Another concern is the risk of over-reliance on Chat GPT, which could weaken users' analytical and critical skills, resulting in passive acceptance of system-generated responses and reinforcing existing biases (Fuchs, 2023). Another risk is 'hallucinations': linguistically correct answers with reference inaccuracies (Ciotti, 2022). Users risk problematic results without knowing AI's limitations, especially when using it like a search engine (Mazzini, 2022).

We decided to focus on the latter aspect. Chat GPT's use as a search engine is notable, as ge-

nerative AI increasingly mirrors traditional search engines, aiding in tasks like text writing, communication practice, issue clarification, and even self-diagnosis (Flores-Cruz, 2023; Shahsavar, Choudhury, 2023). GPT's ability to generate misinformation through content that appears reliable has already been demonstrated, exposing users to new forms of misinformation for example in medicine or even in generating references for research papers (Spitale et all, 2023; Haider et all. 2024). Particularly relevant is the lack of users' habit to fact-cheking information provided by GPT (Grotlüschen et all, 2024). Therefore, the problem is widespread and global and can also extend to historical information, which until know is underrepresented in research on this subject. A related and important question is: if Chat GPT and other generative AIs produce texts that appear reliable but are often inaccurate or distorted upon closer examination, what impact does this have on the public perception of history?

McLean, Roberts, and Gibbs' study highlights the potential of AI in generating historical narratives but also exposes significant limitations. AI introduced biases, oversimplifications, and factual inaccuracies, known as "hallucinations." While the generated biographies were coherent, they often distorted historical realities by following common narrative structures like the hero's journey. This underscores the need for human oversight to maintain accuracy. While both this article and the study address concerns about AI-generated content, McLean, Roberts, and Gibbs focus on evaluating AI's use with pre-verified data (McLean, Roberts, Gibbs, 2024), whereas this article examines its autonomous use by the general public.

We asked Chat GPT about Italian Fascism. This study examines 84 questions submitted to each of the three active versions of GTP (3.5, 4.0 and 4o) for 252 answers and a corpus of 136,450 words. A single account submitted the questions in Italian. A new chat was opened for every question to avoid the risk that GPT could have adjusted the replies because of perceived feedback. By operating in this way, it is possible to measure the AI's original capacity to generate responses about Italian Fascism.

The results are also affected by the fact that there is no public knowledge of the sources used to train Chat GPT. Moreover, different versions were trained on different bases: 3.5 had a much smaller dataset than 4.0 and 4o, which share the same dataset. Finally, GTP's current access to books is still limited by copyright protection (Cristianini, 2024)[1]. Therefore, one must also bear in mind that different models of Chat GPT can yield varying outcomes, and newer versions do not always provide better content. In our case, while GPT-4o and 4.0 claim that Fascism's welfare system was public, GPT-3.5 disagrees, aligning more with academic consensus was a mixed and incoherent system (Pavan, Giorgi, 2022).

The replies have been collected and analysed according to the following categories: factual errors; lack of chronological context, that is, when there is not a single date in the reply; bibliographic inaccuracies; and ambiguous interpretations, which require at least some basic historiographical knowledge to deconstruct them. The analysis is conducted in two parts: quantitative, measuring the number of errors and their nature, and qualitative, offering examples based on the dataset of the complexity necessary to deconstruct replies.

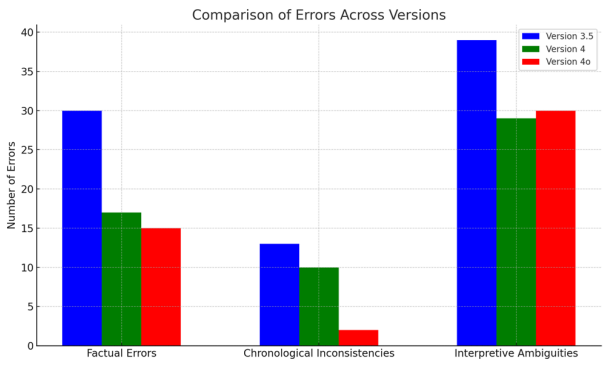

Chart 1. Error recurrence in collected replies divided per GTP version. Source: Responses generated by ChatGPT (versions 3.5, 4 and 4o) in the context of the study described.

The dataset and use of the Italian language impact the findings. Indeed, ChatGPT's responses can vary depending on the language of the prompt (Fleisig et al., 2024), but for our research, the use of Italian is particularly significant. This choice aligns with our objective to demonstrate how GPT reproduces misleading narratives and stereotypes embedded in Italian public memory. The persistent dominance of an opposition between historiography and the public discourse—which sustains an outdated understanding of fascism despite notable advancements in historical research—underscores a concerning gap between academic knowledge and public perception (De Bernardi, 2018). As we will illustrate, the use of ChatGPT further exposes users to this issue.

The issues emerging from this experiment raise broader questions about historical information in the digital age, particularly users' exposure to incorrect information that is presented as reliable and, therefore, capable of influencing their views

on history, but also reflects GTP tendency to indulge with national stereotypes in describing historical phenomena.

Notes

1. GPT-3 operates on approximately 500 gigabytes or 500 billion words and has 175 billion parameters, whereas GPT-4 has around 1.6-1.8 trillion parameters and a dataset of 13 trillion tokens. However, in the case of GPT-4, these figures are estimates, as the exact dimensions are not disclosed.

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